

Topological sectors carry their own temperature

A proposal: a machine-checked median theorem, a torus vortex thermometer with its known answers,
the data that can test the hypothesis, and what would kill it

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SocrateAI-Scientific-QuantumFluids

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Abstract

A pre-registered intervention on a two-dimensional Bose gas [1] found that two boxes with the same energy, particle number and momentum, one carrying four injected vortex pairs and one carrying the same energy as phonons, read different temperatures on the same thermometer, the vortex box being colder by 5–16%. We propose the reading that the literature on point-vortex matter has held since Onsager [2, 3] and that two experiments made quantitative [4, 5]: the vortex configuration is a subsystem with its own temperature, weakly coupled to the phonon bath, and a thermometer that reads the bath reads one of two temperatures. We state the hypothesis in a form that can fail, prove its combinatorial core in Lean 4 (the inverse temperature of a state that mixes sectors is a median of the sector inverse temperatures, so equal energy does not imply equal temperature once a conserved label exists), build the instrument that measures the second temperature on a torus (the Weiss–McWilliams point-vortex energy with its published known answer, and a canonical Monte Carlo calibration after Groszek et al.), and list the data that can test the hypothesis: our own runs with two thermometers, the public vortex-position data of Gauthier et al., and the superfluid-density data of Christodoulou et al. Every prediction below is fixed before the corresponding run or analysis. No physics is claimed as new.

1 The observation and the hypothesis

Observation (round 2 of [1]). On three equilibrated base states, arm V (four imprinted vortex–antivortex pairs) and arm P (the same ΔE as random phonons, matched to machine precision) were compared over $t \in [100, 300]$. The condensate fell by 0.49–0.73 in V and by at most 0.035 in P; the coherence exponent grew 9–44 $\times$ in V; and the two-window equipartition thermometer read V colder than P by 5.5%, 9.2% and 16%, and colder than the untouched base by 4%, with total energy conserved to 2×10^{-7} . The pre-registered composite claim required equal readings and was not made.

Hypothesis F. The microcanonical entropy depends on the sector, $S(E, N, P, W)$, so $1/T = \partial S / \partial E$ at fixed W is sector-dependent. Concretely the field is a two-temperature system: a phonon bath at T_b and a vortex gas at $T_v = (\partial S_v / \partial E_v)^{-1}$ defined on the vortex positions [2]. Topological protection is the weak coupling that keeps $T_v \neq T_b$ for a long time. Injecting a configuration with $T_v \gg T_b$ (an imprinted arrangement near the maximum-entropy energy of the vortex gas) draws energy from the bath into the vortex system as the configuration relaxes towards its own equilibrium and nucleates further pairs; the bath cools.

What would kill it. (K1) The torus vortex thermometer, calibrated canonically, fails its closure test (set β , read β back within 10%). (K2) On the V arm, T_v and T_b agree within their errors from the first block on. (K3) The bath's energy loss (kinetic energy at $k \geq 0.4 k_{\text{cut}}$) is not matched, within 30%, by the gain in low- k incompressible flow energy of the field (Nore–Abid–Brachet decomposition) over the same blocks – the field functional, not the point-vortex Hamiltonian, whose per-pair additive constant makes cross- N budgets void. (K4) On the Gauthier data, the decay time of the clustered configuration does not fall as the condensate fraction falls.

2 The combinatorial core, machine-checked

Let Ω be a finite set of microstates with energy level $\varepsilon : \Omega \rightarrow \mathbb{N}$ and sector label $w : \Omega \rightarrow \mathbb{Z}$. Write $n_W(E)$ for the number of microstates in sector W at energy E and $N(E) = \sum_W n_W(E)$. The discrete inverse-temperature factor of a sector is $e^{\beta_W} = n_W(E+1)/n_W(E)$ and the global one is $e^\beta = N(E+1)/N(E)$.

Theorem 1 (`SectorTemperature.globalRatio_mem`). *If every occurring sector has $n_W(E) > 0$ and $r_1 \leq e^{\beta_W} \leq r_2$ for all W , then $r_1 \leq e^\beta \leq r_2$.*

The proof is the mediant inequality for ratios of nonnegative sums; the file also proves $N = \sum_W n_W$, $\log n_W \leq \log N$, equality of the global and sector factors when a single sector is populated, and – via the conservation theorem of [1] – that the sectors are invariant sets of every continuous evolution without a phase slip, which is what makes $S(E, W)$ meaningful. Eight theorems, standard axioms, three negative controls fail (mediant without positivity; global equal to an arbitrary sector; the entropy inequality reversed). Comparator: accepted by the `nanoda` kernel and by the Lean kernel (2026-09-25).

What the theorem says for the observation: a thermometer that does not know the sector reads a weighted mean of sector temperatures, and two states of the same energy in different sectors need not share it. What it does not say: which temperature the equipartition estimator reads on a continuum field. That is an empirical question, and Section 4 is how it is answered.

3 The instrument and its known answers

Torus point-vortex energy. Weiss and McWilliams [6], eq. (12): on the 2π box, $h(x, y) = \sum_m \ln[(\cosh(x - 2\pi m) - \cos y) / \cosh 2\pi m] - x^2/2\pi$, $H = -\sum_{i < j} \kappa_i \kappa_j h(\mathbf{r}_{ij})$, zero total circulation and zero vortex momentum. Physical units here: $E_{\text{phys}} = \pi E_{\text{WM}}$, $\beta_{\text{phys}} = \beta_{\text{WM}}/\pi$ ($\rho = 1$, $\kappa = 2\pi$). Known answers run before any reading: the dipole energy grows by $2 \ln 2$ per doubling of the separation (1.384, 1.377 measured; two-dimensional Coulomb); the density of states of six vortices at zero momentum has rms 2.985 against their 2.90; the mean differs by an additive convention that a temperature cannot see.

Canonical calibration. Metropolis sampling at set β with pair moves that conserve the vortex momentum, for $N = 4$ to 16 vortices, giving $\langle E \rangle(\beta)$ and the dipole and cluster fractions of Groszek et al. [7]. Two internal checks: $d\langle E \rangle/d\beta = -\text{Var}(E)$ at every interior grid point, and closure (K1). **Results** ('vortex_thermometer_canonical.py', 4000 sweeps per β , nine β values from -0.6 to 3): closure passes at all six sizes – set 1.2, read 1.223, 1.313, 1.256, 1.270, 1.281, 1.142 for $N = 4, 6, 8, 10, 12, 16$ (largest deviation 9.4%). A lighter run (1500 sweeps) had failed closure at $N = 6$ and 8 (1.385, 1.334): the failure was precision, not the instrument, and both runs are kept on record. The fluctuation identity holds to 10–30% at $\beta \geq 0.3$ (coarse finite differences), but **fails for** $\beta \leq 0$: $\langle E \rangle$ is flat from $\beta = -0.6$ to 0 where $-\text{Var}(E)$ predicts a slope of -5 to -70 . The pair-move Metropolis does not equilibrate the clustered, negative-temperature branch in these chains (cluster moves are known to be needed there). Consequence, fixed now: the thermometer is valid for $\beta \geq 0.3$ only; no negative temperature is read from it.

First calibrated reading (post hoc, one snapshot at $t = 1500$). Arm V at $e = 0.90$: $T_v = 2.15$ ($N = 12$) and 3.21 ($N = 10$) against bath readings 0.39 and 0.38 in the same block – $T_v/T_b = 5.5$ and 8.4 . Arm V at $e = 0.60$: the imprinted configuration sits at $\beta_{\text{WM}} \approx -0.1$, inside the unvalidated branch; we report $|\beta_v| \lesssim 0.1$, i.e. $T_v \gg T_b = 0.08$, and no sign. The untouched and phonon arms carry 4–6 tightly bound thermal pairs whose energy lies below the coldest calibrated point (separations at the core scale, outside the point-vortex model): no reading. These are the numbers P1 will test block by block; they are consistent with it and do not establish it.

First placement (post hoc, from the round-2 final states). The imprinted configuration at $e = 0.60$, untouched for 1500 time units, sits 0.27σ below the mean energy of random neutral configurations: $\beta_v \approx 0$, next to a bath at $\beta_b \approx 10$. The thermal arms sit $4.5\text{--}6\sigma$ below (bound pairs). This placement is robust; the numerical T_v from the slope of the density of states is not, and is not quoted there; the calibrated reading above supersedes this placement.

4 Data that can test the hypothesis

4.1 Our own runs: two thermometers (pre-registered here)

Re-run the three V arms of round 2 saving vortex positions and signs every $\Delta t = 10$. Predictions: (P1) $T_v > T_b$ in the first block on every base, by more than the combined errors; (P2) $|T_v - T_b|$ decreases monotonically across blocks (Spearman $\rho < -0.7$ on 15 blocks); (P3) the bath’s cumulative energy loss over $[0, 300)$ equals the field’s low- k incompressible gain within 30% (post hoc on the round-2 final states: -50 vs $+63$ at $e = 0.60$, i.e. 26%; at $e = 0.90$ the surplus dominates and the block-resolved run is required); (P4) on the P arms T_v is defined only intermittently (few vortices) and, where defined, agrees with T_b within 20%.

4.2 Gauthier et al. 2019 (Zenodo 2548958, CC BY 4.0)

Unsigned vortex positions at eleven hold times, ten shots each, for six condensate fractions (18–75%). The paper’s Fig. 4 is a known answer: the nearest-neighbour distance ℓ/ℓ_0 relaxes to the uniform value with a time constant that falls as the condensate fraction falls. In this programme’s terms that is link 3’s lifetime: the coupling of the protected configuration to the bath sets its decay. Prediction (P5): recomputing τ from the raw centroids gives a monotone decrease with decreasing condensate fraction, with $\tau(75\%)/\tau(18\%) > 3$. **Result** (run 2026-09-25, ‘exploration/external/gauthier2019_lifetime.py’): $\tau = 6.36(64), 5.41(54), 3.59(77), 3.46(53), 3.46(61), 2.85(114)$ s for condensate fractions 75, 67, 44, 33, 26, 18%. Monotone non-increasing: **yes**. Ratio $\tau(75\%)/\tau(18\%) = 2.2$: the pre-registered threshold **3 fails**. The direction is the paper’s; the size of the effect in this fit (free initial value, relaxation to the uniform value 1) is smaller than the threshold I set. Reported as such.

4.3 Christodoulou et al. 2021 (Apollo 10.17863/CAM.66056, CC BY 4.0)

Superfluid phase-space density $D_s = n_s \lambda^2$ against D/D_c from first and second sound, at $\tilde{g} = 0.64$. Not a two-temperature test, but the external anchor for the stiffness jump this programme located at $T_{\text{BKT}} = 0.821$ in its own units: prediction (P6) the experimental D_s at the last point below D_c lies within 1.5 of the universal value 4, and our ladder’s $n_s \lambda^2$ against T/T_{BKT} overlays theirs against D_c/D within 25% over the common range, once both are plotted against the reduced variable. **Result** (‘exploration/external/christodoulou2021_overlay.py’, Fig. 1): part one **passes** – at $D/D_c = 1.00$ and 1.01 the measured D_s is $2.99(52)$ and $4.93(47)$, both within 1.5 of 4. Part two **fails as written**: five of the six superfluid points overlay our curve within 0.2–14% ($D/D_c = 1.93$: 11.84 vs 11.81 ; 1.51 : 9.10 vs 9.01 ; 1.18 : 6.55 vs 6.48 ; 1.10 : 5.96 vs 5.32 ;

1.01: 4.93 vs 4.22), and the sixth, the point on the jump itself at $D/D_c = 1.000$, is 2.99(52) against our 3.99 – a 33% deviation, 1.9σ of their error bar, where the two finite systems round the jump differently. With no free parameter, a classical field at $g = 1$ and an atomic gas at $\tilde{g} = 0.64$ lie on one curve away from the jump; at the jump the criterion I fixed was too tight for the experimental error, and the verdict stands as fail.

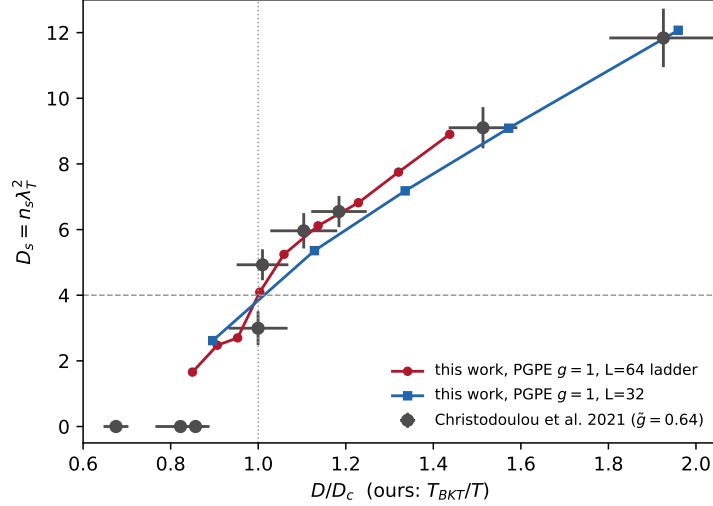


Figure 1: Superfluid phase-space density against the reduced variable: Christodoulou et al. (points with errors) and this programme’s classical-field ladder and finite-size box, with D/D_c identified with T_{BKT}/T . No parameter is adjusted.

4.4 Thouless pumps as the existing matched pair

Lohse et al. [8] and Nakajima et al. [9]: a winding cycle pumps one quantum per period, a non-winding cycle of the same drive pumps none. Cited as the paradigm; no new analysis.

5 Round 3: the two thermometers, block by block

Pre-registered in PGPE_R3_PREREG.md before any run; verdicts in PGPE_R3_RESULTS.md.

Part A (P1–P4). The round-2 arms V and P were re-run on the three bases with vortex positions saved every ten time units, and T_v read block by block through the calibrated thermometer (closure passed at every size used, $N = 4\text{--}40$: reads 1.14–1.31 for set 1.2). **P1 passes on all three bases:** in the first block $T_v/T_b = 7.5$ and 5.6 at $e = 0.90$, and at $e = 0.60$ every sample sits in the hot, unreadable branch ($\beta_v < 0.3$). **P3 passes on all three:** the bath’s loss against the phonon arm equals the field’s incompressible gain within 15%, 7% and 26% ($-57/+67$, $-51/+55$, $-41/+55$). **P2 fails as written:** T_v drifts from 2.9 to 1.9 over 1500 time units while T_b stays at 0.37–0.41, but with a block-to-block scatter of ± 0.4 the Spearman coefficient is -0.56 and -0.47 against the pre-set -0.7 ; the sign is the predicted one on both readable bases. **P4 fails on one base:** the phonon arm’s incompressible energy, a small fluctuating baseline of 25–31 against 55–85 for the vortex arm, moved by 19% where 15% was allowed. The statement that P1 and P3 support, with our own instrument: the injected label carries a temperature five to eight times the bath’s, and the bath’s cooling is the transfer of its energy into the label’s flow field. The rate of their approach is not established.

Part D and an accidental sector experiment. Sixteen pairs were imprinted to test the conservative coarsening exponent of Groszek and Billam; the pre-registered power law was not found on the one clean arm (D1 fails), while the bath cooled below its base value and then heated as pairs annihilated on all three (D2 passes: the cycle that Part A opened closes). The generator had enforced $\sum \mathbf{q}\mathbf{r} \in L\mathbb{Z}^2$ – enough for a single-valued phase – but not $\sum \mathbf{q}\mathbf{r} = 0$, and so imprinted, on two arms, a *net winding around the torus*. Recorded as an amendment after the runs and reported for what it is: on one arm a winding-3 persistent current that is intact after 1500 time units, with 29.5% of the atoms in the mode $k = (-3, 0) \cdot 2\pi/L$ and a $k = 0$ “condensate fraction” of 0.003 because the whole condensate flows, and an incompressible-energy plateau of $248 \approx 178 + 70$ (the current plus the vortices); on the other a winding-2 current that decayed to zero by vortices crossing the torus – the phase slips of `slip_of_winding_change` – heating the bath by ≈ 85 units against the 79 its energy predicts. The ring thought experiment of [1], both branches, observed by mistake on our own torus; single runs, not pre-registered.

Part C ($L = 128$): an admission failure, honestly reported. Six runs at $e \in \{1.00, 1.10, 1.20\}$, two seeds each, energy drift $\leq 1.1 \times 10^{-6}$. The isotropy admission built for $L \leq 64$ ($R_L \leq 1.25$, $R_T \leq 1.1R_L$) rejects four of six: their transverse current fluctuations exceed the longitudinal ones by a factor 2–4, giving an unphysical negative $K = n_s \cdot 2\pi/T$. Block-by-block T_b , condensate fraction and vortex number are flat across the sampled window in every run, admitted or not, so this is not a transient the window missed but a standing feature of the state reached – most likely because the warm-up window ($t_{\text{tr}} = 3000$), carried over unchanged from $L \leq 64$, does not let the larger box’s additional long-wavelength modes equilibrate. On the two admitted points ($e = 1.10$: $\eta K - 1 = 0.284$; $e = 1.20$: $\eta K - 1 = 0.449$) the duality offset is *larger* than at $L = 64$ (0.12, 0.20) – the opposite of the shrinking Hasenbusch trend the criterion was written to see – but with one surviving seed per energy and no admitted point at $e = 1.00$, this is reported, not trusted: it is as consistent with a biased subsample (admission keeps whichever seed happened to isotropise) as with a real finite-size surprise. No crossing of $K = 4$ is bracketed in the admitted range. **C1: fails as pre-registered; C2: not bracketed; both void pending a longer equilibration window at this box size in a future round** – not executed here, since each run already costs ≈ 14 h.

6 Duality is a structure; the sector is the cause

This programme began with an involution, $R \leftrightarrow \alpha'/R$ on a length, and was told that its physical home was T-duality, K3 mirror symmetry, the quantum Hall modular group. What the rounds since have established is where such a duality lives and what it can and cannot do. We state it on the compact boson, whose sectors are the pairs $(n, w) \in \mathbb{Z}^2$ of momentum (spin-wave) and winding (vortex) charge, with $\Delta(n, w; R) = n^2/R^2 + w^2R^2/4$ in the normalisation where the self-dual radius is $\sqrt{2}$ (`CompactBoson.lean`, twelve theorems, three negative controls):

1. **The duality is a reindexing of the sector lattice.** $\Delta(n, w; R) = \Delta(w, n; 2/R)$, and the sector sum $\sum_{\mathbb{Z}^2} F(\Delta, s)$ is invariant under $R \mapsto 2/R$ because the map $(n, w) \mapsto (w, n)$ is a bijection – `Equiv.tsum_eq`, with no convergence hypothesis. Savit’s review [10] gives the general form: a duality is a Fourier transform on the group of sectors, electric charges in G against magnetic charges in $\hat{G}$; the partition function is invariant and no sector is distinguished except at the self-dual point.
2. **The product of the electric and magnetic dimensions does not depend on the radius:** $\Delta(1, 0) \Delta(0, 1) = \frac{1}{4}$. In the superfluid dictionary $\Delta(1, 0) = \eta/2$ and $\Delta(0, 1) = n_s \lambda_T^2/2$, so this is $\eta n_s \lambda_T^2 = 1$: the relation measured to 12–27% on eleven energies in two boxes (Section 5 and [1]). *Our numerical test of the duality was the measurement of this product.*

3. **The self-dual point is not the transition.** $\Delta(1, 0) = \Delta(0, 1) = \frac{1}{2}$ at $R = \sqrt{2}$; the vortex becomes marginal, $\Delta(0, 1) = 2$, at $R = 2\sqrt{2}$, where $\eta = \frac{1}{4}$ (Nelson–Kosterlitz). The transition is the threshold of one sector, not the fixed point of the exchange.

The same triad – a group of sectors, a duality that reindexes them, physics that happens when a sector crosses a threshold or is imposed by an intervention – recurs across domains, and the criterion for when a self-dual point is physical follows from it. Kramers–Wannier self-duality [11] pins the Ising T_c because order and disorder operators exchange and $\mathbb{Z}_2$ has a single symmetric point: the dual sectors are simultaneously critical and the fixed-point structure is discrete. The compact boson has a line of fixed points, and BKT is the marginality of the vortex sector alone: the self-dual point is one point on the line and nothing happens there. In string gas cosmology [12, 13] “the symmetry we make use of is T-duality, and the new degrees of freedom are the winding modes”: momentum modes prevent contraction, winding modes prevent expansion, the radion is stabilised at the self-dual radius as the minimum of a potential in which both sectors are populated – and the exit from the Hagedorn phase is the *annihilation of winding modes*, a sector event. For the superconductor–insulator transition, boson–vortex duality [14, 15] predicts a universal resistance at a self-dual critical point; the cause is vortex (or charge) proliferation, and self-duality constrains the critical point only if the fixed point happens to be self-dual. In the quantum Hall system $\Gamma_0(2)$ organises the plateau diagram, Laughlin’s gauge argument on the Chern sector causes the quantisation, and the Fricke element is not a symmetry of the plateaux (QHFricks.lean).

That is why one involution appears in helium, Hall bars, K3 moduli and string gases, and why it explains none of them. What acts is the sector: conserved by continuity (TopologicalProtection), additive in scale (ScaleResolvedWinding), read as the continuum degree (ContinuumWinding), carrying its own temperature (SectorTemperature), and – in the energy-matched intervention of rounds 2 and 3 – a difference-maker. The duality is exact, verified, and not a cause.

7 What this proposal does not claim

- Novelty of the two-temperature picture: it is Onsager’s, and it is measured [4, 5, 7]. Novelty of the mediant inequality: none; its value is that it is the exact statement, checked by two kernels, of why the round-2 thermometer clause was wrong in principle.
- That the equipartition estimator reads T_b exactly; this is what P1–P4 test.
- Any transfer outside this system without its own matched intervention [16].

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